SWYNEX INTERNSHIP

TASK 1

Data Cleaning and Preprocessing Report

Olist E-commerce Dataset

Project detail	Description
Tools	Python, Pandas, Jupyter Notebook, VS Code
Input / output format	CSV
Datasets processed	9
Status	Cleaning, CSV export and initial readability audit completed

Prepared as a project summary based on the recorded notebook results and validation outputs.

1. Project Overview

The objective of this task was to inspect, clean and prepare the Olist E-commerce dataset for further analysis. The project involved nine datasets containing information about customers, orders, products, sellers, payments, reviews, product categories and geographical locations.

Python and Pandas were used to examine data quality, identify potential inconsistencies, create data-quality flags and export the processed datasets.

2. Tools and Technologies

Tool	Purpose
Python	Data processing
Pandas	Data cleaning and manipulation
Jupyter Notebook	Executing and documenting the workflow
VS Code	Development environment
CSV	Input and output data format

3. Dataset Overview

Dataset	Original records	Final records	Final columns
Customers	99,441	99,441	8
Orders	99,441	99,441	17
Order Items	112,650	112,650	10
Products	32,951	32,951	12
Reviews	99,224	99,224	8
Sellers	3,095	3,095	5
Payments	103,886	103,886	8
Category Translation	71	71	3
Geolocation	1,000,163	1,000,163	8

4. Cleaning Operations Performed

4.1 Customers Dataset

- Checked customer identifiers and missing values.
- Verified duplicate customer IDs.
- Examined location variations using customer_unique_id.
- Created multiple_zip_codes_flag, multiple_cities_flag and multiple_states_flag.
- Preserved all original records.

Location variation results:

Variation	Unique customers
Multiple ZIP codes	250
Multiple cities	122
Multiple states	39

The final dataset contained 99,441 records and eight columns, with no missing values or duplicate customer IDs.

4.2 Orders Dataset

- Examined missing order timestamps and checked the sequence of order-related dates.
- Identified missing timestamp fields and date-sequence anomalies.
- Created quality flags instead of deleting potentially useful records.

Finding	Count
Missing approval timestamps	160
Missing carrier handover timestamps	1,783
Missing customer delivery timestamps	2,965
Delivered orders with missing date information	8
Approval/carrier timestamp sequence anomalies	1,358
Carrier/delivery timestamp sequence anomalies	23

4.3 Order Items Dataset

- Checked exact duplicate rows and uniqueness of the (order_id, order_item_id) combination.
- Examined suspicious shipping limit dates, price values and freight values.
- Created anomaly flags and retained zero freight values for further review.

Finding	Count
Suspicious shipping limit dates	4
Records flagged for unusual prices	8,427
Records flagged for unusual freight values	12,134
Zero freight values	383

4.4 Products Dataset

- Checked duplicate product identifiers.
- Examined missing catalogue information, dimensions and product weight.
- Created quality flags.

Finding	Count
Missing catalogue information	610
Missing dimension information	2
Zero product weight	4

4.5 Reviews Dataset

- Checked exact duplicate rows and repeated review identifiers.
- Examined missing review titles and messages.
- Created duplicate_review_id_flag.

Finding	Count
Repeated review IDs (unique IDs)	789
Records carrying repeated review IDs	1,603
Missing review titles	87,656
Missing review messages	58,247

Missing review text was retained because written feedback may be optional.

4.6 Sellers Dataset

- Checked missing values and seller ID uniqueness.
- Investigated ZIP prefixes associated with multiple states.
- Created multiple_states_zip_flag.

17 ZIP prefixes were associated with multiple states, and 45 seller records were flagged for multiple-state ZIP prefix associations.

4.7 Payments Dataset

- Checked missing values and exact duplicates.
- Examined negative and zero payment values, installment values and payment types.

Finding	Count
Zero payment values	9
Zero installment values	2
Undefined payment types	3

No missing values, exact duplicates or negative payment values were identified. Potentially unusual records were retained for further investigation.

4.8 Category Translation Dataset

- Checked missing values, duplicate records and blank category mappings.
- Verified unique Portuguese and English category values.

The dataset contained 71 records, 71 unique Portuguese category values and 71 unique English category values. No missing or blank mappings were identified.

4.9 Geolocation Dataset

- Examined geographical coordinates and coordinate validity.
- Investigated coordinate pairs associated with multiple cities or states.
- Created geographical quality flags.

Finding	Count
Records flagged outside approximate Brazil bounds	42
Records associated with coordinate pairs linked to multiple states	24
Records flagged for coordinate pairs associated with multiple cities	90,093

The original 1,000,163 records were preserved. These flags indicate records requiring further geographical investigation; they do not independently prove that a coordinate is incorrect.

5. Data Quality and Validation

- Checked whether all nine exported CSV files existed.
- Loaded each CSV file using Pandas.
- Compared resulting dimensions with expected dataset dimensions.
- Checked missing values and duplicate identifiers where applicable.
- Preserved original records when anomalies did not justify deletion.

Final audit result: All nine cleaned CSV files were found and loaded successfully during the final readability audit.

6. Output Files

- olist_customers_cleaned.csv
- olist_orders_cleaned.csv
- olist_order_items_cleaned.csv
- olist_products_cleaned.csv
- olist_reviews_cleaned.csv
- olist_sellers_cleaned.csv
- olist_order_payments_cleaned.csv
- olist_category_translation_cleaned.csv
- olist_geolocation_cleaned.csv

The files were saved in the project's notebook directory.

7. Key Learning Outcomes

- Importing and inspecting CSV datasets.
- Using Pandas for data manipulation.
- Identifying missing values and duplicate records.
- Checking identifier uniqueness.
- Investigating date and geographical inconsistencies.
- Creating data-quality flags.
- Preserving original records during anomaly analysis.
- Exporting cleaned datasets to CSV.
- Validating exported files.

8. Conclusion

The SWYNEX Internship Task 1 involved cleaning and preprocessing nine Olist E-commerce datasets using Python and Pandas. Potential data-quality issues were investigated, appropriate flags were created, and original records were preserved. All nine processed CSV files were exported and successfully loaded during the final audit.

The resulting datasets are prepared for the next stage of the project: exploratory data analysis (EDA) and business insights generation.

Project Status: Data cleaning and CSV export completed.

Next Stage: Exploratory Data Analysis (EDA).